SAMPLE

# The Wave-Particle Duality

Is It a Wave or a Particle?

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## Introduction

Here is the strangest fact in the whole of physics, stated plainly: light, which the nineteenth century proved beyond doubt to be a wave, can strike a detector one whole hit at a time, like rain on a roof; and an electron, which everyone knows to be a particle, a tiny ball of charge with a countable position, can be sent through two slits one at a time and paint, over hours, the striped interference pattern of a wave that went through both. Neither statement is a metaphor and neither is an approximation. The two behaviors are the extremes of one reality that classical language cannot hold, and the physics of that reality is called wave-particle duality, the subject of this book: the discovery that the universe's fundamental entities are neither waves nor particles, and must be described, in different experiments, as each, while being fully neither at any time.

The book tells the story whole, as this series does: the century of light's wave proofs and the particle facts that broke them; Einstein's photoelectric heresy and de Broglie's audacious reversal, matter itself granted waves; the double-slit experiments, for light, for electrons, and finally for whole molecules, each arrival a single click and each accumulation of clicks a striped interference pattern; Bohr's complementarity and the Copenhagen reading; the which-way experiments and the quantum eraser, where the pattern's fate is decided by information, not force; the quantum-field-theory resolution, the modern trade's dissolution of the paradox into something deeper; the interpretations' honest disagreements; and the working machines, microscopes and sensors, built on the electron's wave. The honest account of what is settled, what is philosophical, and what is engineering is the spine of the telling, and the reader who finishes will have the century's central quantum fact in plain language: nature's smallest things answer the question you ask, and if you ask a wave question you get a wave answer, if you ask a particle question you get a particle answer, and nothing in between, because there is no in between.

One more honest note before the story starts, because this subject's popular literature is the noisiest in physics. The double-slit experiment has been called the only mystery, the eraser has been sold as time travel, and the observer has been turned into a mind, none of which the experiments say. This book keeps to what the apparatus shows: stripes and dots, toggles and correlations, whole arrivals and statistical patterns, and where the trade itself is still arguing, the argument is labeled as argument. The wonder needs no inflation, and the reader who wants the difference between the quantum facts and the quantum folklore will find it marked in every chapter.

## Chapter 1: The Discovery: Two Centuries of Certainty, Broken Twice

The story properly begins with a war over light that ran for two hundred years and was, the textbooks say, settled twice. Newton, the seventeenth century's greatest authority, held light to be a stream of particles, corpuscles flying from the source in straight lines, and his authority carried the day for a century; Huygens, the Dutch contemporary, held light to be a wave, and lost the first round to Newton's reputation. The reversal came at the turn of the nineteenth century: Thomas Young's double-slit experiment, light passing through two narrow openings and painting dark and bright stripes on the far wall, stripes that no stream of balls could ever paint, together with Fresnel's mathematics and Maxwell's crowning identification of light as an electromagnetic ripple, settled the war by 1900 in the wave's favor, and the settlement was total: light was a wave, the nineteenth century's physics knew it, and the knowledge was the crown of its optics.

The break came from an unglamorous corner, and the reader should feel how small the crack looked. The photoelectric effect, light knocking electrons out of metal, behaved wrongly in every wave-theory detail: the electrons came out instantly or not at all, with energies that depended on the light's color and not at all on its brightness, while the wave theory, which should have let any color's waves shake electrons loose if the shaking were strong enough, predicted delay, and brightness-dependence, and got neither. Einstein's 1905 explanation, light delivered in indivisible packets whose energy is set by the color, explained every wrong detail at a stroke, and the trade resisted it for two decades, the wave's evidence being a century deep, until the experiments, and the work of the decade that followed, forced the impossible conclusion: light, the wave of waves, arrives in whole units. The discovery of the duality, in its modern form, was complete when de Broglie reversed the trick in 1924, granting to matter, by symmetry, the waves light had lost, and the electrons confirmed it within three years; the reader should hold the shape of the whole discovery: two centuries of certainty, broken twice, once for light and once for matter, by the same mathematics in both directions.

A note on the word quantum, which this book will use as the founders did. The word means amount, the smallest amount, and it entered physics as an accounting term: light's energy arrives in amounts, whole units, no fractions delivered. The century's discovery, compressed into the one word, is that the universe's smallest bookkeeping is digital: energy in whole packets, arrivals in single events, the possibilities continuous but the facts countable, and the duality is the visible face of the whole account, the wave being the continuous possibility's picture, the particle the countable fact's. The reader who finds the word dry may keep its provenance: the driest accounting word in the language was drafted, by accident, to name the century's deepest surprise, and the surprise made the word into the name of an age.

The discovery chapter owes the reader one more scene, the first solvay-era confrontation, because the duality's founding arguments were public. The 1927 Solvay meeting, the quantum generation's gathering, staged the debate that this book's interpretation chapter inherited: Einstein proposing, over breakfast, thought experiments to break the new bookkeeping, Bohr answering each from the complementarity's grammar, the pattern of attack and defense establishing, in front of the trade's whole generation, that the wave and particle accounts were experimentally untouchable and philosophically contested, the meeting's participants dividing into the readings that persist today. The reader should hold the scene's honest meaning: the duality was never a settled textbook fact quietly absorbed; it was argued over, in public, by the trade's best, from the first, and the arguments' descendants are still being had, the subject's second century opening with the founders' questions still open, which this book has reported as it stands.